



SnapSeek 2.0 at Video Browser Showdown 2025

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Abstract. SnapSeek 2.0 is a novel system designed for information retrieval from videos participating in the Video Browser Showdown 2025. The system aims to enhance interactivity and accelerate query input speed by employing a human feedback procedure and integrating a magic brush for automatic sketch completion. Building on SnapSeek, which debuted at LSC 2024, the new version retains its predecessor's most notable features to boost performance including sequencing of scenes in videos and expanding metadata through a variety of computer vision and NLP techniques. To adapt to the competition, specialized datasets including MVK and LapGynLHE undergo detection and segmentation processes, making retrieval more efficient and accurate. Additionally, the system employs a minimalist interface to ensure user-friendliness.

Keywords: Interactivity · Multimedia retrieval · Scene sequencing · Sketch-based retrieval

1 Introduction

In today's world, the demand for querying video information is growing, but this task is challenging due to the vast and diverse range of video data across multiple fields. The Video Browser Showdown (VBS) [12] was established to address these challenges, running for over 12 years to find the most comprehensive systems for video retrieval. Recently, VBS has expanded its datasets to include MVK, focusing on marine videos, and LapGynLHE, featuring endoscopic gynecological data, further diversifying the sources for video information retrieval.

In this research, we present the SnapSeek 2.0 system, an advancement derived from the original SnapSeek system [4], which was initially introduced at Lifelog Search Challenge (LSC) 2024. This iteration of SnapSeek incorporates several novel contributions:

1. SnapSeek 2.0 enhances the efficiency of query search processes through the implementation of an innovative user interaction technique, the Magic Brush.

2. Enhancing the exploration and accuracy of the system by leveraging the video scene sequencing feature and extending metadata through the implementation of scene text recognition, image captioning, transcript speech-to-text, along with object, color, and human pose detection.
3. A deeper understanding of MVK and LapGynLHE datasets by exploring objects in the video frames using image segmentation and object detection techniques.

2 Related Work

In the VBS 2024, the *Visione* system [2], developed by Giuseppe Amato and colleagues, demonstrated a significantly more user-friendly interface. Furthermore, *Visione* incorporates a variety of advanced AI models, enhancing its capability to analyze video data effectively. The system’s encoding of objects and colors for object positioning has also yielded impressive results. Consequently, these innovative features have propelled *Visione* to become the champion of VBS 2024.

AVSeeker by Tu-Khiem Le et al. [7] at VBS 2022 introduces a novel search approach where the system actively suggests relevant concepts to refine retrieval results, instead of passively providing paginated candidates based on user input. It proposes a temporal concept enrichment process, integrating action/activity recognition to enhance concepts extracted from video clips. Additionally, it features a query-by-sketch mechanism, allowing users to retrieve images by sketching scenes, addressing limitations in visual concept detection.

In the most recent edition of VBS, VBS 2024, Gia-Huy Vuong et al. proposed ViewInsight [17], a chat-based video retrieval system that enables users to enhance their queries or refine the ranked results through multiple rounds of interactive, context-based chat with the system. Their system also offers a user-friendly temporal query feature, allowing users to search for results that include temporal information in the query.

3 Data Pre-processing

3.1 Dataset

The competition uses three datasets. The Vimeo Creative Commons Collection (V3C) [11] includes 28,450 videos (3,800 h) across three parts - V3C1, V3C2, and V3C3 - and provides metadata, shot-segmentation, and keyframes. It has been a core part of VBS since 2019, with V3C3 released in 2024. The Marine Video Kit (MVK) [14], introduced in 2023, contains 1,300 underwater videos from 36 global locations, posing challenges due to its unique characteristics and domain-specific content. Lastly, the LapGynLHE dataset, featuring laparoscopic gynecology videos, debuted in VBS 2024, also requiring advanced retrieval techniques due to its specialized medical context.

3.2 Basic Techniques

We leverage several techniques to enhance the use of the V3C dataset. First, **scene text recognition** is employed using the multilingual PaddleOCR¹ model, enabling keyword-based searching for more relevant image retrieval. Additionally, inspired by Ribeiro et al. [9], we integrate contextual descriptive captions using the BLIP-2 model to improve **image captioning**. Lastly, **transcripts**, generated via the Whisper Speech-to-Text API from Rossetto et al. [10], provide crucial context for understanding video content.

3.3 Advanced Techniques

3.3.1 Encoding Objects, Colors, and Human Pose

Inspired by the method proposed by Amato et al. [1], we develop a module for efficient retrieval based on objects, colors, and human poses, utilizing drag-and-drop and sketching interactions.

We use YOLOv8 [16], trained on the OpenImagesV7 [6] dataset to detect objects because of the vast number of classes it has been trained on (up to 601). The output of object detection models is bounding boxes so to encode objects more precisely, we use Segment Anything Model (SAM) [5] to generate segmentation masks. Using these masks, we determine if an object is present in a given cell. For colors, we followed Amato et al. [1] to detect the dominant colors of the cells in the grid and store their encoding. For human pose detection, we utilize the YOLOv8-pose model [16] to detect 17 keypoints corresponding to body parts.

Deviating from Amato et al. [1], we overlay a 20×20 grid on the image to encode objects, colors, and pose keypoints while using the same descriptor as them. Once the encodings are generated, they are indexed and made searchable using Elasticsearch. The user interface features a whiteboard equipped with tools such as a brush for drawing color or object regions, a tool for creating bounding boxes for objects and colors, and a tool for placing human poses. These visual concepts are then encoded into textual form and searched in Elasticsearch's indexes.

3.3.2 Pre-processing for Specialized Datasets: MVK

To effectively retrieve information from the MVK datasets, the Magic Brush feature is employed along with the additional data about marine species' position in the images. In a study by Truong et al. [14], the MVK dataset showcases raw and unprocessed data about many marine species. Therefore, we first filter out blurry images caused by motion or obstacles, then normalize the images to align them in a consistent direction. Next, we employ three models-EMOD [13], FishInv, and MegaFauna²-to detect the species. To further assist in searching among the numerous labeled organisms, we develop a biological dictionary, based on Visual

¹ <https://github.com/PaddlePaddle/PaddleOCR>.

² <https://github.com/Orange-OpenSource/marine-detect>.

KIS, that helps users identify species from the species names, descriptions, or images.

3.3.3 Pre-Processing for Specialized Datasets: LapGynLHE

To enhance the metadata of LapGynLHE videos, we focus on surgical tool recognition, which serves as both an object filtration method and a component of our system’s Magic Brush feature. Based on Sanat Ramesh et al.’s ablation study [8], we employ the Moco v2 SurgSSL model which is trained on the Cholec80 dataset [15]. The model can recognize seven tools: grasper, bipolar, hook, clipper, irrigator, scissors, and specimen Bag, allowing precise tool detection and positioning. After retrieving the bounding boxes for the surgical tools, their locations are encoded using the same method in Sect. 3.3.1 before feeding to Elasticsearch Fig. 1.

4 System

4.1 An Overview of SnapSeek 2.0 System

4.1.1 User Interface

Snapseek 2.0’s UI is designed with a strong focus on functionality and user experience, particularly in freely customizing image display for users’ convenience. It emphasizes usability, making it easy to perform queries and quickly browse results.

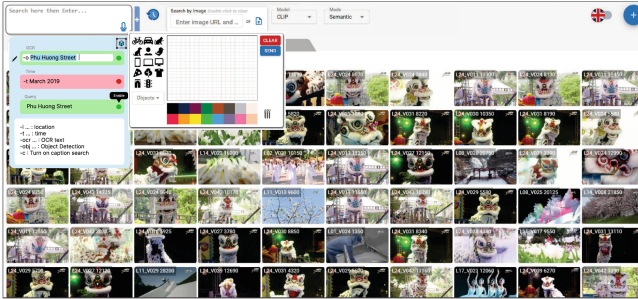


Fig. 1. SnapSeek 2.0’s user interface.

Customizable Image Display. SnapSeek 2.0 dedicates over 80% of its interface to image display, enabling users to view multiple images simultaneously. Furthermore, it also offers customizable settings, allowing users to adjust the image grid, pop-ups, and whiteboard layout for a more personalized and intuitive experience. Additionally, users can customize the number of feedback requests to match their preferences, enhancing overall usability.

Frontend-based filtering with Fuse.js. SnapSeek 2.0 utilizes Fuse.js³ for frontend filtering, which enhances performance by minimizing the load on the backend server. When users query data, the relevant metadata described in Sect. 3.2 is also fetched from the backend. Fuse.js enables fuzzy search on this metadata using the Bitap algorithm, allowing users to simply input a filter category and value into the search box for filtering. This client-side filtering delivers faster response times and instant feedback during searches, making it especially effective for large datasets.

Multi-language support via Google Cloud Translation API. Snapseek 2.0 supports queries in any languages using Google Cloud Translation API.

4.1.2 Search Functionalities

SnapSeek 2.0 is a multimodal image retrieval system offering flexible, customizable search options that combine various query modalities, including:

- **Semantic Text Search:** OpenCLIP ViT-H/14 and BLIP2 ViT-g models are used for cross-modal searching in a unified text-image vector space, with Milvus as the vector database.
- **Image Search:** Users can input an image file, clipboard-cropped image, or image URL to find visually similar results, ideal for visual KIS tasks.
- **Keyword Search:** Text data from OCR, descriptions, and transcripts are summarized and indexed into Elasticsearch for fast retrieval of relevant frames via exact or fuzzy queries.
- **Encoding Search:** This feature allows users to sketch or specify visual concepts, including objects, colors, and human poses, for more visually-focused searches (see Sect. 3.3.1).

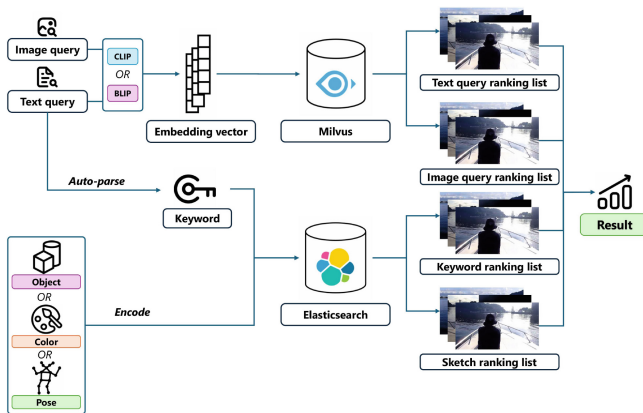


Fig. 2. The Overview of SnapSeek 2.0 searching functionalities.

³ <https://www.fusejs.io/>.

4.2 Scene Sketching with Automatic Completion

In certain instances when sketching a complex scene, it is necessary to fill a region with a specific color to search for more precise frames, which also requires the shape the user is drawing to be closed-shape. To accelerate this process, we have developed an automatic completion tool that can quickly fills the area within a contour drawn by the user Fig 2.

If the contour remains open after drawing, the tool connects the start and end points using Bresenham’s line algorithm [3]. It then identifies a bounding box around the contour, scans it, and selects a point inside by detecting crossings of the contour an odd number of times. From this point, a Breadth-First Search (BFS) fills the enclosed region.

This method assumes simple, non-intersecting shapes, as the algorithm only fills the first closed region detected. Despite this limitation, the approach ensures high processing speed.

4.3 Scene Sequencing in Videos

This version of SnapSeek builds on the activity sequence from LSC’24, adapted for the V3C dataset. V3C videos consist of distinct scenes with changes in camera angle, focus, or characters, while frames within each scene remain visually consistent. To segment the list of video’s keyframes, we detect content changes by comparing consecutive keyframes, using a similarity threshold. Instead of low-level features like color histograms, we encode keyframes using the OpenCLIP ViT-H/14 model, which provides a more comprehensive semantic understanding.

To facilitate viewing, we developed a “Timeline” tab that displays videos segmented by scenes. Each scene is represented by a group of three keyframes that illustrate the sequence of the scene. We also added an expand button, allowing users to view the full set of keyframes for each scene, making it easier to observe the complete story at a glance.

4.4 Human Feedback

During development, we noticed relevant images were scattered in the search results, often appearing alongside irrelevant or placeholder images like scene transitions or single-color frames, which reduced the search experience. To address this, we added a feedback feature that lets users “like” or “dislike” images. As more feedback is collected, the system improves at promoting relevant images, ensuring they appear earlier in the results.

Specifically, each feedback request includes lists of relevant and irrelevant images, called “original” images. For each “original” relevant image, we identify k similar images, then rerank these candidates based on the “original” image’s score and similarity to the “original” image, forming the A^+ set. Similarly, we construct the A^- set for irrelevant images.

After each feedback round, the set A is updated, prioritizing elements from A^+ based on descending similarity. The update formula is displayed below. This

iterative user feedback process continuously enhances the accuracy of image retrieval, improving the search and exploration experience.

$$A \leftarrow (A \cup A^+) \setminus A^- \quad (1)$$

5 Conclusion

In summary, SnapSeek 2.0 is a versatile multimodal information retrieval system capable of handling tasks such as text-to-image and image-to-image searches. We've enhanced metadata auto-parsing by incorporating image captions, transcripts, and details about objects, colors, and human poses. Additionally, the system introduces advanced features like scene sketching with automatic completion and scene sequencing for videos. For specialized datasets like MVK and LapGynLHE, we leverage pretrained models to optimize their performance for these applications.

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